Recitation 9

ID	Age	Income	Freq	Points	Class
1	25	50	8	120	Y
2	45	80	3	50	N
3	28	55	9	150	Y
4	50	85	2	40	N
5	30	60	10	180	Y

KNN, K = 3

ID: 1, $x = (25, 50, 8, 120)$

d from ID: 2

$$d = \sqrt{(25-45)^2 + (50-80)^2 + (8-3)^2 + (120-50)^2}$$

$$= \sqrt{400 + 900 + 25 + 4900}$$

$$= \sqrt{6225} = 78.9$$

d from ID: 3

$$d = \sqrt{(25-28)^2 + (50-55)^2 + (8-9)^2 + (120-150)^2}$$

$$= \sqrt{935} = 30.6$$

d from ID = 4

$$d = \sqrt{(25-50)^2 + (50-85)^2 + (8-2)^2 + (120-40)^2}$$

$$= \sqrt{8286} = 91.0$$

d from ID = 5

$$d = \sqrt{(25-30)^2 + (50-60)^2 + (8-10)^2 + (120-180)^2}$$

$$= \sqrt{3729} = 61.1$$

ID	distance (d)	class
3	30.6	Y
5	61.1	Y
2	78.9	N
4	91.0	N

K = 3, nearest

ID 3, 5, 2 = (Y, Y, N) Majority = Y

Actual = Y, Predicted = Y

— Iteration 1 —

ID = 2, (45, 80, 3, 50) = x

d from ID = 1

$$d = \sqrt{(45-25)^2 + (80-50)^2 + (3-8)^2 + (50-120)^2}$$

$$= \sqrt{6225} = 78.9$$

d from ID = 3

$$d = \sqrt{(45-28)^2 + (80-55)^2 + (3-4)^2 + (50-150)^2}$$

$$= \sqrt{10950} = 104.6$$

d from ID = 4

$$d = \sqrt{(45-50)^2 + (80-85)^2 + (3-2)^2 + (50-40)^2}$$

$$= \sqrt{151} = 12.29$$

d from ID = 5

$$d = \sqrt{(45-30)^2 + (80-60)^2 + (3-10)^2 + (50-180)^2}$$

$$= \sqrt{17574} = 132.6$$

ID	distance (d)	Class	, K = 3 Nearest
4	12.29	N	4, 1, 3
1	78.9	Y	(N, Y, Y)
3	104.6	Y	
5	132.6	Y	Predicted = Y

Actual = N $\neq$ Predicted = Y

Wrong prediction

Cross Validation Metrics

* Mean CV Accuracy

$$\bar{x} = \frac{1}{2} = 0.5 = 50\%$$

* Standard deviation

$$\sigma = \sqrt{\frac{1}{N} \Sigma (x_i - \bar{x})^2}$$

$$\sigma = 0$$

* 95% confidence level

$$CI = \bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

$$= 50 \pm 1.96 \cdot 0/\sqrt{1}$$

$$= 50 \pm 0$$

$$\Rightarrow (50\%, 50\%)$$